

a compact vector signal generator (VSG) N5186A MXG with four channels, enabling signal generation up to 8.5GHz and providing 960MHz of modulation bandwidth per channel. It caters to the demanding requirements of dense wideband multichannel applications by offering multiple, individually intricate signals. The signal generator simplifies complex



setups, reduces external connections, and accommodates up to four

channels within its compact 2U form factor. It features an integrated reflectometer to ensure the precise delivery of signals to the device under test. The vector signal generator delivers dependable and reproducible results, making it ideal for various commercial and aerospace defense applications. It uses custom DAC application-specific integrated circuits and direct digital synthesis to produce precise signals, minimising distortion and aligning with evolving benchmarks for component and module design. The integrated reflectometer streamlines setup procedures by optimising the device under test's match, resulting in accelerated time-to-test.

Keysight Technologies

<https://www.keysight.com/in/en/home.html>

Radar testing systems for autonomous vehicles

Rohde & Schwarz has launched the R&S AREG800A automotive radar echo generator and the R&S QAT100 antenna array. Elevating radar testing innovation to new heights, this system enables rigorous testing of radar-based advanced driver assistance systems (ADAS) and radar sensors integral in automated driving cars, entirely over the air, by simulating comprehensive driving scenarios. The revamped radar test system (RTS) serves as the tool for radar object simulation, encompassing

single sensor trials to immersive 360° environments, tailored for the critical 76-81GHz bandwidth of automotive radar sensors. The RTS promises scalability and adaptability, facilitating a wide range of electronically generated ADAS traffic scenarios throughout



the testing lifecycle. The synergy between the backend powerhouse of these two devices, used in benchtop development, hardware-in-the-loop, and vehicle-in-the-loop applications, is at the core of this revolution. The company claims that this system streamlines the transition from real-world test drives to laboratory settings, anticipating errors at an early stage and resulting in significant cost savings, advancing the industry with foresight and expertise.

Rohde & Schwarz

<https://www.rohde-schwarz.com/in>

3D automated X-ray inspection

Test Research, Inc. (TRI) has introduced the TR7600 SV series, an advancement in 3D automated X-ray inspection (AXI) technology. Boasting a 7µm resolution for precision



defect detection and higher yield rates, this series incorporates advanced artificial intelligence (AI) algorithms that outperform conventional grey-level-based methods, ensuring accurate void defect identification. Equipped with an EtherCAT motion controller for real-time measurements and simplified maintenance, this AI-enhanced inspection system aligns seamlessly with Industry 4.0 standards, facilitating integration with manufacturing execution systems. Beyond rapid image reconstruction and flaw identification, these inspection tools

stand out by safely securing samples on the conveyor during examination, thus preventing additional damage.

Test Research, Inc.

<https://www.tri.com.tw/en/index.html>

Advanced 4-channel oscilloscope

UNI-T has introduced the 4-channel digital storage oscilloscope series UPO1000. The series of digital oscilloscopes utilises digital 3D technology, fast acquire phosphor. Offered in bandwidths of 50MHz, 100MHz, and 200MHz, they boast a real-time



sampling rate up to 2GSa/s. Standard across the series are four channels,

the inclusion of an independent DVM module, and a plethora of trigger and bus decoding features, including full-memory hardware real-time decoding capabilities. The versatile oscilloscope series is beneficial across a wide array of sectors such as communication, computing, IC design, instrumentation, industrial electronics, consumer electronics, automotive electronics, field maintenance, and research & development/education.

UNI-T

<https://instruments.uni-trend.com>

EMBEDDED

SoM and starter kit

AMD expands its footprint in the industrial sector with the introduction of the Kria K24 system on module (SoM) and the KD240 drives starter kit. The SoM encapsulates powerful computing capabilities in a compact format, specially designed for budget-conscious industrial and commercial edge applications. With its low latency features, it plays a crucial role in powering electric drives and motor controllers, particularly in digital signal processing applications at the edge. Its primary application areas include